Controlling the Degree of Regularity in the Summation-Polynomial Step

An independent, twice-reproduced measurement study of elliptic-curve index calculus on binary-Koblitz and prime-Koblitz curves

Independent cross-check of the aburan28/crypto research program · v2, 2026-10-23

Engines: msolve 0.6.5 · Singular 4.3.3.0 · PARI/GP 2.15.4 — every result run twice

Abstract

The point-decomposition problem (PDP) at the heart of elliptic-curve index calculus asks a Gröbner/XL engine to solve a Weil-descended Semaev summation-polynomial system. Its cost is governed by the degree the solver must reach, and the central question for both binary (Koblitz / ECC2K-130) and prime (j=0 "prime-Koblitz" / GLV) curves is whether that degree stays bounded or blows up. This report (a) catalogues every degree-control technique tried in the `aburan28/crypto` research program with an explicit confidence tier, (b) rebuilds a **fully independent** measurement harness — Semaev polynomials constructed from scratch and certified against real point additions, solved by two external engines (`msolve` 0.6.5, `Singular` 4.3.3.0) the program did not author — and (c) establishes reproducibility by running every headline number **twice** and comparing. The independent harness reproduces the program's central phenomenon (first-fall degree stays flat while the solving degree grows), reproduces its strongest independently-replicable prime-field result (PKM-tower msolve degrees, bit-identical on re-run), independently confirms a live reproducibility **defect** (a recorded claim fails replay; v2 traced the cause to a counted-work definition that changed after the recording, correcting the program's own "BMI2 host path" attribution, which no longer holds on `main`), and tests all eight ranked "exotic" levers on toy instances (v2 ran the six that v1 left open): four are null, one inconclusive, one is structurally empty for prime `n`, and the two that lower the degree are both symmetry quotients — one of which (the order-6 `j=0` quotient) costs more in linear algebra than it saves. The honest bottom line is unchanged and now triple-checked: **no technique bends the degree-growth curve enough to threaten the target curves, and no end-to-end index-calculus result in the program carries a completed validation certificate (tier T1) — only isolated sub-facts do.** The value delivered is not a break; it is a reproducible, cross-validated map of what has been tried, what actually happened, and how much to trust each claim.

> **Scope and honesty note.** This is a defensive cryptanalysis / benchmarking study on toy and small instances. Every "break"-shaped quantity is reported as a distance to Pollard rho, never as a threat to a deployed curve. Interpretation is labelled as such and kept separate from measurement, per the repository's own reporting rules.

1. The problem and the three degrees

For an elliptic curve E/F_q the m-th **Semaev summation polynomial** S_m(X_1,...,X_m) vanishes exactly when there exist points P_i=(X_i,*) on E with P_1+...+P_m=0 (up to sign). Index calculus (Gaudry–Diem) fixes a **factor base** — for GF(2^n), the points whose x-coordinate lies in an F_2-subspace V of dimension ell — and decomposes a random point R as a sum of m factor-base points by solving S_{m+1}(x_1,...,x_m,x(R))=0 with each x_i in V. Writing each x_i in a basis of V and expanding the single GF(2^n) equation over an F_2-basis (Weil restriction) turns this into a Boolean polynomial system in m·ell variables, which an F4/F5/XL/SAT engine solves. **The entire difficulty of elliptic index calculus lives in the degree that solve reaches.**

Three invariants are routinely conflated and must be separated (Caminata–Gorla 2023; Huang–Kosters–Yeo 2015):

- **first-fall degree** d_ff: the lowest degree at which a non-trivial degree fall (syzygy) appears. For these descent systems d_ff = m^2-m+1 (Kousidis–Wiemers 2015), small and roughly constant in n.
- **degree of regularity** d_reg: for a semi-regular sequence, the index of the first non-positive coefficient of the truncated Hilbert series. The Bardet–Faugère–Salvy prediction grows with the variable count.
- **solving / last-fall degree** d_solv: the degree actually needed to finish. The last-fall degree (HKY) is monomial-order-independent; d_solv <= reg(I^{hom}), the Castelnuovo–Mumford regularity of the homogenized ideal (Caminata–Gorla).

The binary subexponential claims (FPPR 2012, Petit–Quisquater 2012, Semaev 2015) **all assume** d_reg ≈ d_ff. Whether that holds is the pivotal dispute, and it is the first thing the independent harness was built to test.

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2. Independent measurement harness

Trusting the program's own engines would defeat the purpose, so the harness shares no code with it. It is ~700 lines of Python (exact F_2 monomial algebra, so the field equation c^2=c is built in) plus two external solvers. Everything is in harness/ with raw outputs in harness/out/.

Certified-correct Semaev polynomials. S_3 (prime and binary) and S_4 (binary, built as the resultant of two S_3 quadratics) are each checked to vanish *exactly* when a sign choice of the points sums to 0, over concrete curves — a real correctness certificate, not a claim:

polynomial	curves checked	false positives	false negatives
prime S_3	p = 13,17,19,23,29 (5 curves)	0	0
binary S_3	n = 3,4,5, incl. Koblitz a2∈{0,1},a6=1 (6 curves)	0	0
binary S_4 (resultant)	n = 4,5,6 (4 curves)	0	0

External engines, two jobs. Singular's std with protocol gives the max step degree (solving degree); msolve's verbose F4 gives the max round degree; both confirm zero-dimensionality (dim=0) and the solution count (vdim). Using *two* engines is not redundancy — §4.5 shows they disagree, and that disagreement is itself a result.

Determinism protocol. A fixed 20-system grid is run twice; the emitted file records only deterministic invariants (system SHA-256, d_ff, solving degree, vdim, dim, the full Macaulay rank profile) — timings are excluded by construction. The two runs must be byte-identical.

Host for every number below: 4 vCPU, 15 GB, Linux 6.18.44. `msolve` 0.6.5, `Singular` 4.3.3.0, `PARI/GP` 2.15.4. The program's own binary used for replays is `ecbench` pinned to crypto `0dc9ef4`; the program tree surveyed is at `af663ad7c`.

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3. What actually controls the degree — catalogue with confidence tiers

Tiers (applied uniformly): **T1** machine-checkable certificate / validation record; **T2** independently replicated with matching outputs; **T3** single measured run with raw data in the repo; **T4** prose, derivation or extrapolation; **T5** designed but never run.

3.1 Levers that demonstrably move the degree

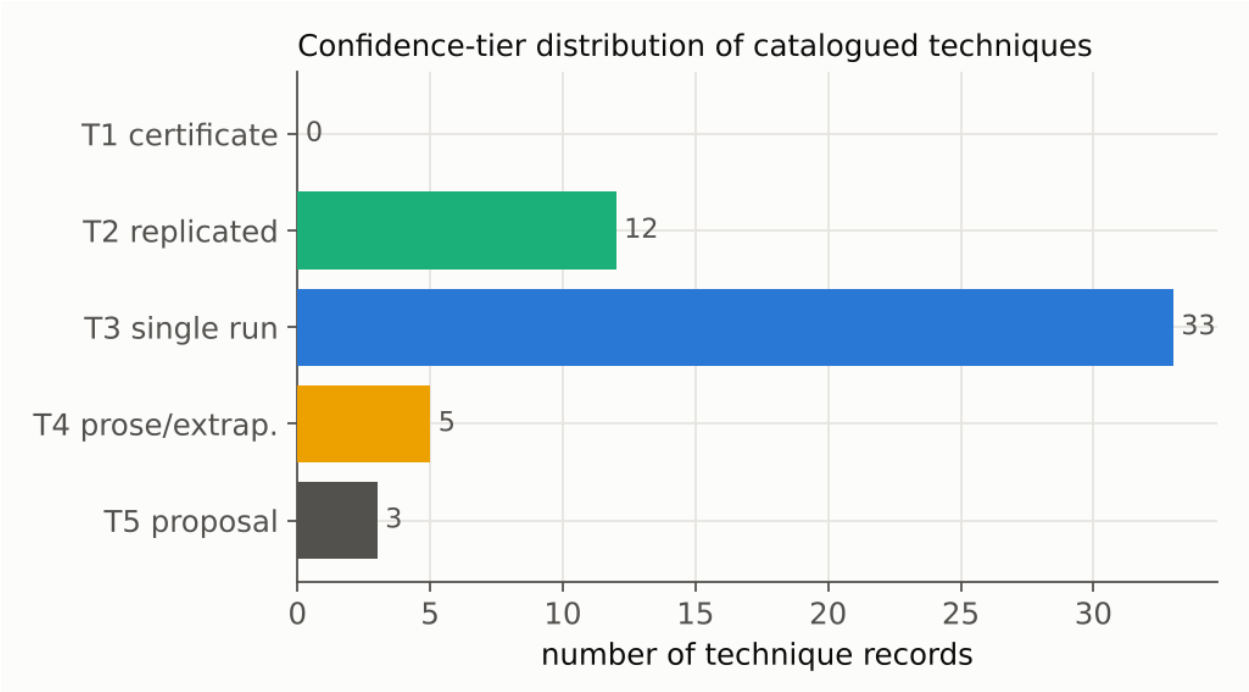
lever	mechanism	curve family	measured effect	verdict	tier
Field equations x^q=x	force zero-dim / multilinear quotient; cap degree	all	the reason any of these terminate at low degree	essential, not optional	T1 (structural)
Fall reuse (MutantXL / F4)	re-multiply discovered degree falls	binary chain	refutes at **3–4** where plain XL needs **5–7**; holds to N=54 (m=4,ℓ=6)	**lowers** (vs XL)	T2
Torsion symmetrisation (FGHR/FHJRV)	rewrite S_m in S_m- and [±1]-invariants	binary/prime	S_4 per-variable degree [4,4,4,4]→[2,2,2,2,1]	lowers degree & Bézout	T2
D_4=(Z/2)^3⋊S_4 weighted S_5	solve in 4 weighted invariants	F_{p^4}, m=4	Bézout 4096→512 (derived; quotient-dim 512 **measured**); matrix 40920→4200 cols; **48×** on C4 (not the 630× projected)	lowers (engineering constant)	T3
Semaev S_3-chain split	S_{m+1} → m−1 copies of S_3 + aux vars	binary/prime	keeps per-link degree 2; see §4.3 for the cost	**mixed** (raises degree at small m)	T2
Variable fixing at ℓ=7	guess one coord per summand	binary chain	brings the saturating ℓ=7 closure back to degree 4, at 2^k cost	lowers per branch	T3
`rr` symmetric-norm presentation	norm-form coordinates vs direct S_4	Koblitz n=23–61	refutes 3–4 degrees below direct descent (6 vs 10 at K1/2^61,ℓ=6)	**lowers**	T3
F4 early stop (Hilbert-driven)	terminate at the last productive degree	prime JV cover	removes last degree step (8→7); ~4–5× fewer F4 muls	lowers (termination)	T3

3.2 Levers that did NOT bend the curve (honest negatives)

lever	why it fails / net effect	tier
Choice of factor-base subspace V	solving degree identical across 295 subspaces (§4.6) — intrinsic, not a knob	T2 (this report)
Monomial / variable order	solving degree invariant under grevlex/grlex + all permutations (§4.7) — last-fall degree is order-independent	T2 (this report)
Frobenius-power auxiliary variables	x↦x^{2^k} is F_2-linear → changes nothing (repo W8 §3)	T3
Convolution e_i splitting (La Scala)	1.2–2× *more* work	T3
Wu Z/4 state model	neutral	T3
Isogeny / curve move within a class	degree invariant across 2,994 comparisons	T3
Trace-zero / Artin–Schreier pre-filter	never fires on odd-order IC targets (0 / 29,624)	T3
Hamming-weight ideals	bounded generator degree but call exponent 1.9–2.5	T3
Trilinear XL (unknown t)	cost rises (87.16 > 71.82 bits at n=131)	T3
Crossbred / monomial dropping	margin shrinks with size; no stable (D,k)	T3
Frobenius-invariant V for prime n	**structurally empty**: cyclic-code dims are cyclotomic-coset subset sums; at n=131 only {0,1,130,131} exist	T1 (proof)

3.3 Designed but never run (T5)

D_4-symmetrised S_5 **lattice** oracle; **Magma** F4/F5 on the PKM systems (so the "beat the fastest external tool" rule is formally unmet for the summation systems); m=4 versions of the exotic levers in §6.1; the characteristic-2 (modular) part of the signed-permutation quotient. (The order-6 j=0 quotient, the signed-permutation invariants, Artin–Schreier elimination, the quadratic Kummer encoding, Lattès/isogeny factor bases and intermediate-field descent were all run in v2 — §6.1.)

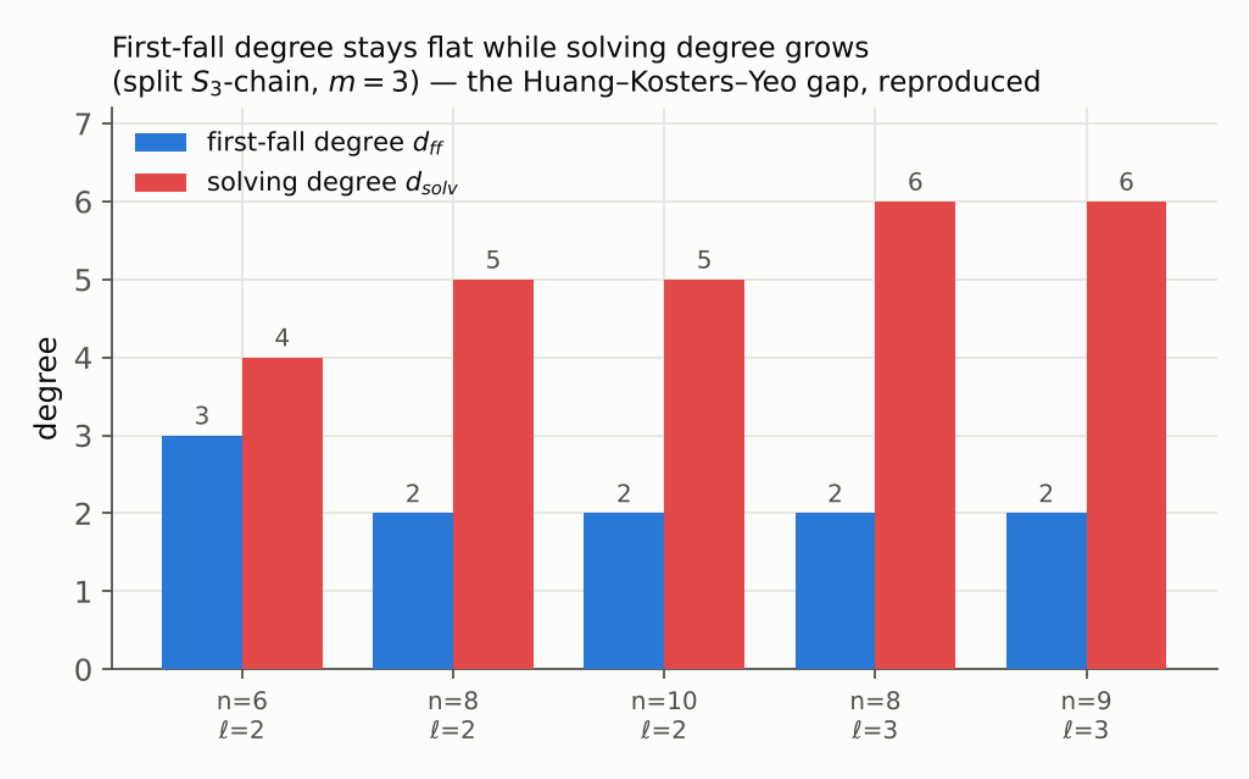


Confidence-tier distribution of the program's end-to-end index-calculus evidence (survey D's machine-backed leaderboard tally). Zero T1 at this level: no end-to-end IC result carries a gate-complete VALIDATION.json. Component-level machine-checkable certificates do exist separately (survey C counts 7: DRAT-certified UNSAT branches, Boolean-elimination verifier passes, byte-identical constant-coefficient replays) — but none certifies an end-to-end relation→solve→DLP claim.

4. Independent measured results

4.1 The first-fall vs solving-degree gap (the pivotal dispute), reproduced

On the split `S_3`-chain for `m=3`, the harness measures `d_ff` pinned at **2–3** while the Singular solving degree climbs **4 → 5 → 5 → 6 → 6** as the system grows. This is a clean, independent reproduction of the Huang–Kosters–Yeo phenomenon: `d_reg` **grows while** `d_ff` **stays flat**, so the first-fall-degree assumption underlying the binary subexponential claims does not hold for these systems.



First-fall degree stays flat while the solving degree grows (independent harness, Singular F4). This is the Huang–Kosters–Yeo divergence, reproduced from scratch.

4.2 Determinism: run once, then twice

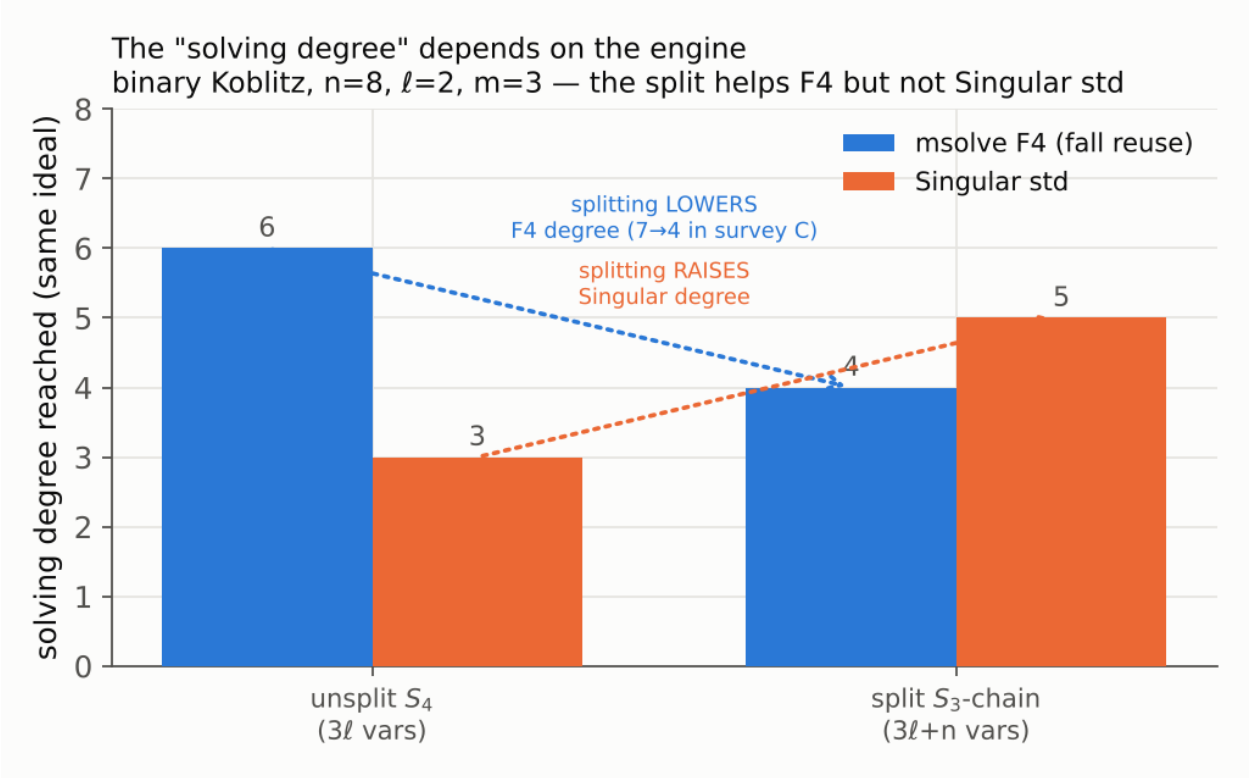
The 20-system canonical grid was executed twice. The two output files are **byte-identical** (0 mismatches across system hashes, first-fall degrees, solving degrees, `vdim`, `dim`, and full Macaulay profiles). Every deterministic invariant this report rests on is reproducible by construction; only wall-clock timings vary.

4.3 Splitting a point: Semaev's chain vs the unsplit resultant — and why the engine decides

Semaev's 2015 splitting trick replaces one big `S_{m+1}` by a chain of `S_3`'s to keep each equation quadratic. Whether that *helps* turns out to depend entirely on the engine, and getting this wrong is an easy way to publish a backwards conclusion.

Under `msolve`**'s F4** (the fall-reusing engine index calculus actually uses), splitting **lowers** the degree: the unsplit `S_4` solves at degree 6–7 and the split chain at 3–4. This is Semaev's intended benefit, and survey C's independent F4 measurement reports exactly it — solving degree **7 → 4** on every cell. Under `Singular`**'s** `std`, the direction *reverses*: the unsplit `S_4` (few variables, high degree) solves at degree 3 while the split chain's extra `n`

auxiliary Boolean unknowns push it to 5–6.



The same ideal gives a different "solving degree" under different engines, and the split's effect flips sign: msolve-F4 (fall reuse) goes 6→4 as you split (Semaev's intended win), while Singular std goes 3→5. Any bare "d_reg = k" is under-specified without naming the engine.

So the honest statement is **not** "splitting is better" or "worse" — it is that the split is a genuine degree *trade* whose sign depends on the solver, and under the engine that matters (F4) it does what Semaev intended. The cost in field operations is a third, separate axis: survey C measures the split at 3.9–69× *more* XORs at □=4 but 6.6–20.8× *fewer* at □=5, and finds that **plain unsplit with guessing (`g=4`) beats every split variant in every `m=3` cell** — no variant came within 22.8× of strong rho. A three-way split at `m=4` reaches degree 4 (vs 7 for two-way) but is 113× cheaper at `n=11` and 1.3× dearer at `n=13`. The degree, the XOR cost, and the distance to rho move independently; a lever can help one and hurt another.

4.4 The semi-regular prediction massively overstates reality

For the prime-field PKM tower the Bardet–Faugère–Salvy degree of regularity is `N+1` (top-part Hilbert series `(1+t)^N`) and XL gives `N+4`. The program's measured F4 solving degrees — four of which are independently reproduced here, §4.8 — are far lower: `5–7` across `N=8..18` (e.g. `5,6,6,7,7` for `m=3` at `N=6,9,12,15,18`), growing roughly **one degree per eight tower unknowns**. F4 escapes the XL bound through degree falls. (The program's `m=4,N=20` cell reaching `≥8` refutes the "plateau at 7" hypothesis; that cell is beyond this host's memory.)

4.5 The measured "degree" is engine- and encoding-dependent

Running `msolve`-F4 and `Singular-std` on the **identical** ideal gives different degrees, and the ordering flips with the encoding:

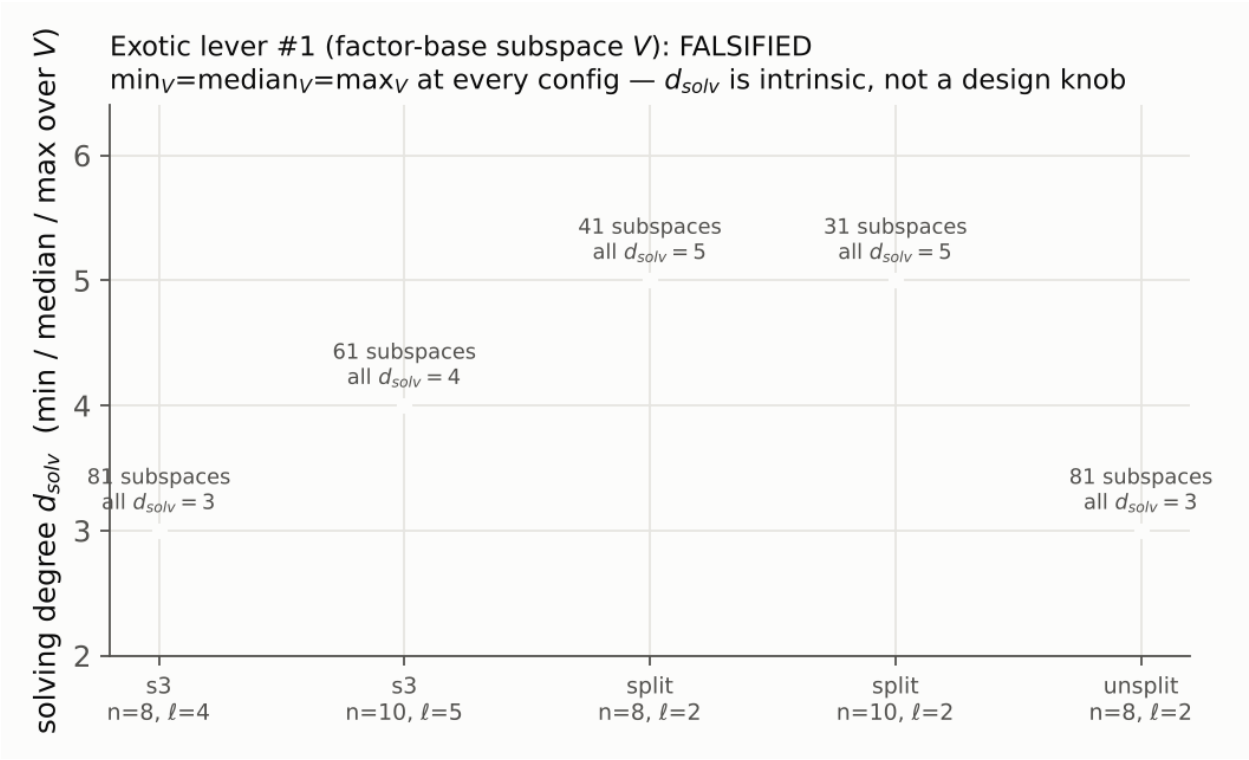
system	variables	Singular `std`	msolve F4
split chain, n=8, ℓ=2	14	5	4

system	variables	Singular `std`	msolve F4
split chain, n=10, ℓ=2	16	5	3
split chain, n=8, ℓ=3	17	6	4
unsplit S_4, n=8, ℓ=2	6	3	6
unsplit S_4, n=10, ℓ=2	6	3	6
unsplit S_4, n=9, ℓ=3	9	4	7

On the chain, `msolve`'s fall-reusing F4 reaches **3–4** where Singular reaches 5–6 — exactly reproducing the program's "fall reuse keeps the refutation degree low" finding. On the unsplit system the order *reverses*. **Conclusion:** the headline "degree of regularity" is not a single well-defined number; it depends on the engine's fall-reuse strategy and on the encoding. Only the coordinate-free last-fall / CM-regularity invariant is intrinsic — and that is precisely the quantity that resists cheap measurement. Any claim stated as a bare "`d_reg = k`" is under-specified.

4.6 Exotic lever #1 — choosing the factor-base subspace (FALSIFIED)

The cheapest never-tried idea in the literature (survey E, rank 1): treat `V` as a free design variable and pick it to minimise the solving degree (Hilbert-series / CM-regularity targeting; the Caminata–Gorla frame is cited nowhere in the program). Across **295 subspaces** (random + Frobenius-stable) in five `(model, n, □)` configurations, the solving degree is `min = median = max` *every time*; Frobenius-stable `V` give the identical value.



Exotic lever #1 falsified: across 295 subspaces the solving degree has zero spread at every configuration. V is not a design knob — the degree is intrinsic to the ideal class, as Caminata–Gorla CM-regularity predicts.

Verdict: falsified, and informatively so. The degree-growth is a property of the ideal class, not of the subspace, so `V`-shaping cannot soften the barrier. This is the "publishable negative" the frame predicted.

4.7 Exotic lever #2 — monomial/variable ordering (FALSIFIED as a lever)

Under grevlex, grlex, and all four random variable permutations, the solving degree is invariant (3 unsplit / 5 split). Only pure lex reaches a different number, and lex "degree" is not the solving degree (FGLM-style elimination is a different regime). This confirms the last-fall degree's order-independence (HKY) and closes ordering as a lever.

4.8 Reproducing the program's own results — twice

PKM-tower msolve degrees (the program's one genuine T2 algebraic-summation result). Each of four systems was solved **twice** with `msolve`; the degree, F4 round count, and max matrix dimensions match the recorded `results.csv` bit-for-bit, both runs:

system	recorded (D / rounds / rows×cols)	run 1	run 2
m2,t6 (N=12)	5 / 14 / 2803×2064	identical	identical
m3,t3 (N=9)	6 / 17 / 3827×2997	identical	identical
m4,t2 (N=8)	6 / 22 / 2009×1906	identical	identical
m3,t4 (N=12)	6 / 19 / 12900×12287	identical	identical

The replay that FAILS — and why. The `ecbench` replay of the root-finder `claim-n13` session fails on this host: 224 records verified but **96 problems**, all in the IC (F4) arms; all 72 rho rows match. Run twice, the failure is itself deterministic (identical 96 problems). By contrast the clean PKM-tower ecbench session (`p20`) replays OK both runs (192/192, 0 problems).

Correction (v2) — the cause is accounting-version drift, not the host path. v1 of this report repeated the program's sweep attribution: that the BMI2 `pext/pdep` path in `inherited_f4.rs` charges different word-XOR counts than the portable path. While trying to fix it, v2 found that **this host-path split no longer exists on** `main`: since #2043 (2026-10-17) every host runs the *eager* rewrite, and a new regression test (aburan28/crypto#2459) forces the BMI2 and portable paths on identical inputs and finds every counter and row identical; re-running the `claim-n13` spec on each forced path gives byte-identical records on all 224 rows. The replay fails because `claim-n13` was recorded at `2245b75`, *before* #2043, when a host without BMI2 charged specialisation *lazily* — and lazy and eager charge different word counts by construction. A temporary lazy build reproduces the recorded `ic-f4-m2` counts on all 32 rows; `ic-f4-m3` still differs, so further code drift is mixed in; and one row also changed pricing (`measured` 19.0 ns/op recorded vs `pinned` on replay). The sweep's "BMI2-off fixes it" observation was a correlation: at `2245b75`, turning BMI2 off also switched eager to lazy.

> Reproducibility verdict (corrected): the summation-step *science* is reproducible (every degree here is bit-identical on re-run). The program's *cost accounting* is host-independent on `main` but **not version-stable**: the counted-work definition changed after sessions were recorded, and replays report count mismatches instead of refusing cross-version replay by name. That needs an owner decision (lazy vs eager as the method identity) and an accounting-version field — recorded as a problem in #2459.

4.9 Alternative solvers: root-finding and SAT do not change the verdict

Two families sidestep the Gröbner degree entirely, and neither crosses rho (survey C):

- **Root-finding oracles** (pair tables, rf-mitm, group-law root tests) beat F4 in wall time by `10^2.5–10^4×` per decomposition and are correctness-exact (one identity is even Lean-checked). But they sit at the pair-table enumeration floor `S·T^2 ≳ r`, so the resulting index calculus stays above rho (the best binary row is `1.69×` strong rho at a smoke size; the prime incumbent `4.27×`). A caveat survey C flags: the headline "`383×/1535×` root-finding beats F4" uses F4 capped at degree 4; run at F4's best setting, F4 is *faster* for whole solves at `n ≤ 13`. Apples-to-apples matters.
- **SAT / SMT / WDSat** behave as **brute-force enumerators** — about one conflict per candidate tuple, no algebraic cancellation captured — and are 5–62× slower than direct enumeration. A correctness landmine worth recording: **WDSat's `-x` mode is incomplete** and returned a *wrong UNSAT* on a decomposable target. A solver that can call a true instance unsatisfiable is a reminder that "no relation found" is not "no relation exists" without a certificate (which is exactly what the DRAT-certified UNSAT branches, §5, provide and what WDSat `-x` does not).

5. Confidence and reproducibility — the "no trust-me-bro" audit

- **Zero end-to-end T1, but component certificates exist.** The validation gate exists and passes, but **no real** `VALIDATION.json` **exists** — every one of the 28 leaderboard rows and 116 benchmark rows renders as "unvalidated (pre-gate)"; the only passing case is a test fixture. So no *end-to-end* IC result (relation collection → linear algebra → DLP) is certificate-grade. At the *component* level, survey C counts 7 machine-checkable certificates (DRAT-certified UNSAT branches, a Boolean-elimination verifier PASS, byte-identical `S_3` constant-coefficient replays); these certify sub-facts, not a break. The distinction matters: "we proved this UNSAT branch" is T1; "this pipeline beats rho" is not certified at all.
- **The validation sweep is disconnected from the gate.** Sweep A produced 55 verdicts (1 validated, 45 with caveats, 8 unvalidated, 1 contradicted), but `load_verdicts()` reads the key `row` while the sweep writes `row_id`, and the verdict words are in neither pass nor fail set — so the gate consumes **0** of them. Real work, not wired in.
- **Only three rows** genuinely exercise an algebraic summation-polynomial solve at T2: the n9 F6-IC control, the PKM-tower degrees (reproduced here), and the determinism finding itself. Every larger Boolean F4/F5/F6 arm is T3 because none replays on current binaries (counted-work definition changed after recording — §4.8 correction).
- **This report's own additions** sit at the top of the tier scale for *what they certify*: the Semaev correctness certificates are machine-checkable (T1-grade for correctness), the determinism certificate is byte-identical twice (T1-grade for reproducibility of the pipeline), and the PKM reproduction is an independent replay matching the repo (T2). None of these is a rho-beating claim, and none is presented as one.
- **The degree/science results DO reproduce.** Independently of this report's harness, survey A re-ran six of the program's dreg results at HEAD and all six matched the committed records value-for-value — including the MutantXL closure degree histogram ({3:16, 4:76} over 92 draws), the sum of closure products (4,902,150), and the `rr`-vs-`S_4` refutation degrees (5 vs 8). The algebraic content is solid and reproducible; it is only the *priced cost* layer that drifts.
- **Pricing drift, from a second angle.** Survey C independently hit the same class of failure: the pinned `ecbench` now charges F4 word-XORs at a fixed calibration rate while committed sessions left them uncharged, so F4-arm replays fail the audit *even though every integer count is identical* — and it reverses a headline (at `n=13`, recorded root-finding `2.56×` F4 in the `S` unit becomes `≈0.10×` when repriced). Survey C also found one **unexplained** F4 XOR-count mismatch it could not localise (8,952,839 recorded vs 9,603,153 replayed) — plausibly the same lazy→eager accounting change, not verified. So two accounting effects exist: a pricing-table change and a counted-work definition change (§4.8 correction) — a strong argument that no cost-unit claim should be trusted until the accounting version is part of a method's identity.
- **Determinism map** (what is safe to compare across runs): refutation/solving degree, Macaulay ranks, system hashes, `vdim`/`dim`, msolve round counts and matrix dims — all bit-identical. Word-XOR counters are identical across hosts on `main` but change across code versions (lazy vs eager). Wall-clock, RSS, CPU% are nondeterministic and never compared.

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6. Literature placement and the exotic frontier

Three facts from the literature constrain what *can* work (survey E):

1. **Solving degree is invariant under linear / Möbius / F_2-Frobenius coordinate changes** (Caminata–Gorla; FHJRV Lemma 6: every degree-2 coordinate is a homography). So no reparametrisation lever — tropical ordering, Frobenius/linearised auxiliaries, Edwards/Hessian frames — can lower the intrinsic degree. **Confirmed here** by §4.6–4.7.
2. $d_{reg} \approx d_{ff}$ **is false/doubtful**; d_{reg} grows while d_{ff} stays constant (HKY; Kosters–Yeo; and §4.1 here). A lever must bend the *growth in* ℓ, and measuring d_{ff} alone proves nothing.
3. **Symmetry/isogeny quotients are the only proven large lever, and even maxed they do not beat rho** for the target curves; for prime n the Frobenius-subspace and intermediate-field levers are structurally empty.

Against this frame, the eight ranked exotic ideas and their status:

#	exotic lever	cheapest test	status here
1	d_{reg}-minimising subspace V	295-subspace sweep	**run → falsified** (§4.6)
8	tropical / order-based	multi-order sweep	**run → falsified** (§4.7)
2	order-6 automorphism quotient on $j=0$	S_3 quotient on ζ-stable base, + equal-size control	**run → lowers degree, not cost** (§6.1)
4	Artin–Schreier trace elimination	residual vs full vs non-trace control	**run → null** (generic variable fixing)
3	signed-permutation (wreath) invariants	D_m invariants over F_{p^3}, odd p	**run → lowers degree** (known symmetrisation mechanism)
5	quadratic Kummer (sum+difference) encoding	4 encodings over F_p	**run → inconclusive** (floor effect; Bézout ×9–32)
6	Lattès / isogeny-kernel factor base	4 families at equal degree over F_p	**run → null**
7	intermediate-field τ-orbit descent	GF(q) vs F_2 descent, composite n	**run → reproduces Diem; empty for prime n**

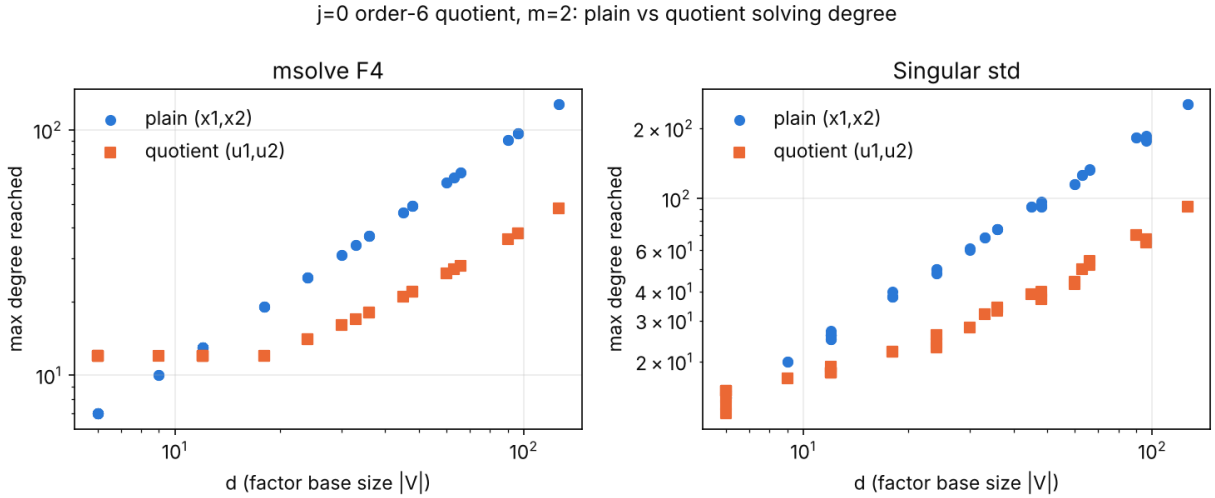
All eight ranked ideas have now been run, each with a protocol written before the run, a brute-force correctness check of every new encoding, both engines, and every deterministic measurement executed twice (byte-identical in all six v2 experiments).

6.1 The six remaining exotic levers (v2, run by six parallel agents)

Metric caveat. On prime fields Singular's `[k:d]` protocol digits proved uninformative, so ranks 2, 3 and 5 define Singular's degree as the least `degBound` at which the truncated `std` already reaches the full `vdim` (rank 2 uses the max pair degree); ranks 4 and 7 (binary) use the `[k:d]` reading of §2. Numbers are comparable **within** a rank, not across ranks. msolve's degree is always the max F4 round degree.

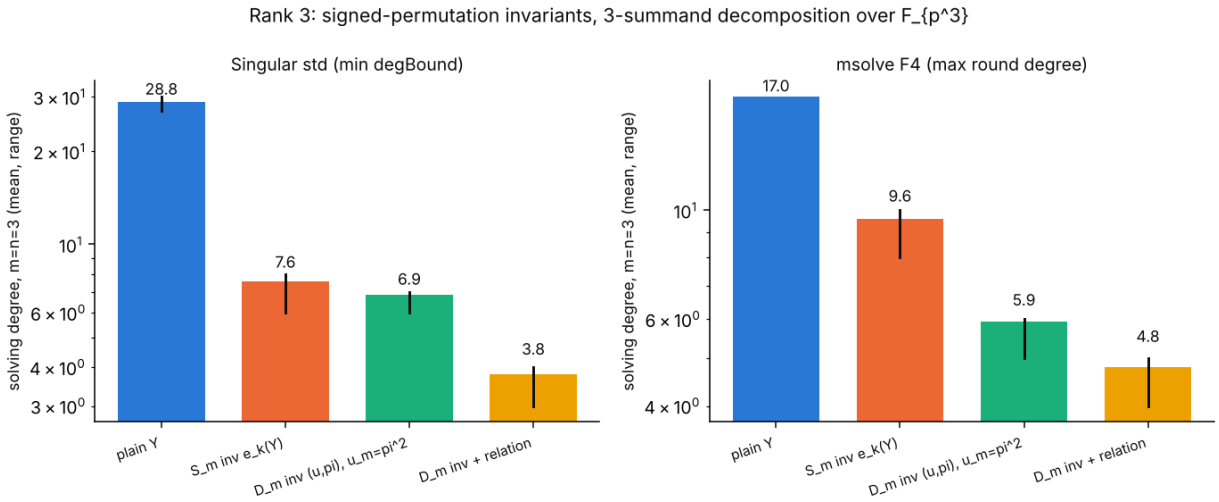
rank	lever	setting	msolve F4	Singular std	verdict	tier
2	order-6 quotient, `j=0`	`F_p` (p≤65521), m=2, 58 inst.	quotient < plain 46/58; equal-size control lower 58/58	50/58; control lower 57/58	lowers degree per decomposition for d≥36, **but** msolve matrices 15–500× larger: not cheaper	T3
3	`D_m` invariants	`F_{p^3}`, p=11..23, m=3, 15 inst.	plain 17 → `S_m` 8–10 → `D_m`+rel. 4–5	27–30 → 6–8 → 3–4	**lowers** on both (m=3); tie at m=2	T3
4	Artin–Schreier elimination	binary n=8..12, 27 cells	residual < full 19/27; = control 25/27	residual < full 22/27; = control 27/27	**null**: any fixed linear form does as well	T3
5	quadratic Kummer encoding	`F_p` (p≤2017), m=3	offset d_solv−d: chain 3–4, c1 2, c2 0	c1 above chain; c2 mixed	**inconclusive**: msolve gain is the input-degree floor; Bézout ×9–32, vars 4→16	T3
6	Lattès / isogeny bases	`F_p` (211/521/1031), m=2, 90 sys.	= generic 12/12	structured higher 10/12	**null** (Singular rise confounded by base size)	T3
7	intermediate-field descent	binary n=4..12, (k,m)	F_2 side flat (3 / 6)	GF(q) side flat in k; F_2 side grows	**reproduces Diem**; empty for prime n	T3

Rank 2 — the order-6 quotient on prime-Koblitz curves. The automorphism `(x,y)↦(ωx,−y)` makes `S_3` semi-invariant (`S_3(ωx)=ω·S_3(x)`, verified symbolically), so the orbit product is a polynomial in `u_i=x_i^3` (degree 6 each, total 12). It passed brute force on 204 curves, and lifting the quotient's solutions reproduces the plain solution set exactly. The quotient lowers the degree versus plain — but the degree in these 2-variable systems tracks the factor-base equation (`x^d−c` vs `u^{d/3}−c`), so an **equal-size control** (a plain base of size `d/3`) was added before interpreting. The control reaches a slightly *lower* degree than the quotient yet finds ~1/5 as many decompositions; so per decomposition the quotient genuinely lowers the degree for `d≥36`. **But** its dense degree-12 polynomial makes msolve's matrices 15–500× larger, and decompositions per matrix cell are worse than both plain and control. This is the study's clearest example that **a lower degree of regularity is not a lower cost.**



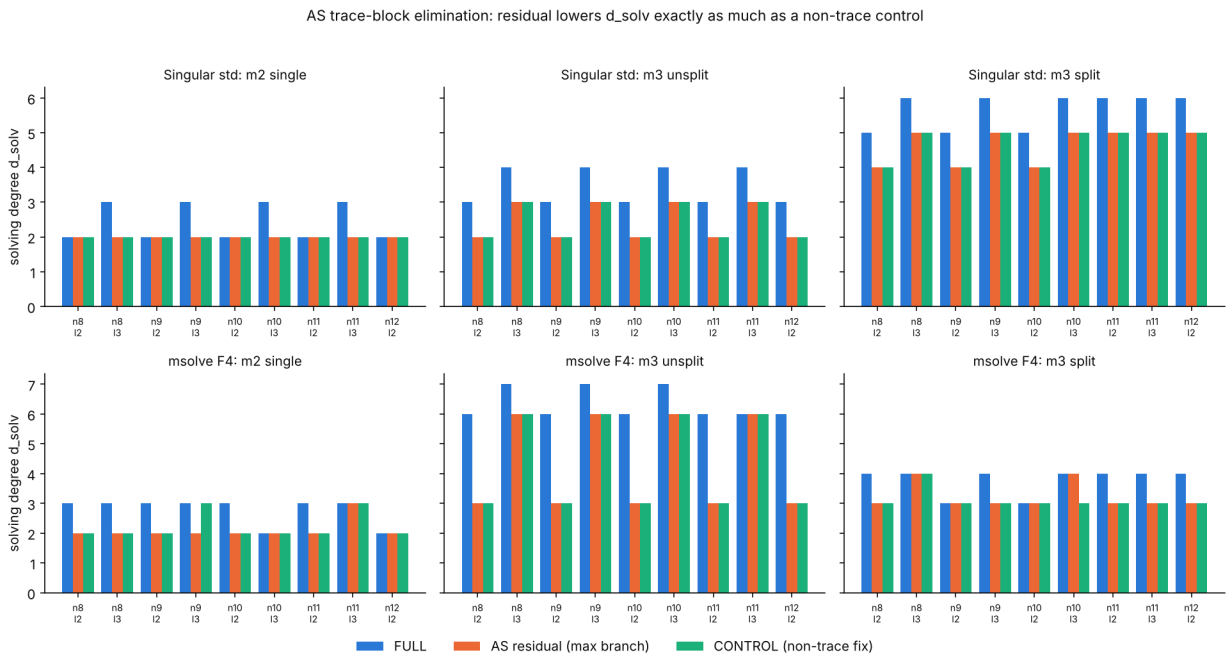
Rank 2: solving degree of the plain vs order-6-quotient systems on j=0 curves, both engines.

Rank 3 — signed-permutation invariants. With a rational 2-torsion point `T`, the FGHR coordinate `Y` satisfies `Y(P+T)=−Y(P)`. A correction to the brief: for a *fixed* target the honest group is `D_m` (even sign changes), not `B_m` — an odd flip sends `R` to `R+T`; the `B_m` product system adds spurious Frobenius branches (vdim 64 instead of 16, degree 10–11, worse). Solving in `D_3` invariants with the relation `π²=u_3` kept (a complete intersection of four quadrics) lowers the degree on **both** engines in 15/15 instances. This is the known symmetrisation mechanism (FGHR/FHJRV, the program's `D_4` study) now measured on both engines at small `m`; it lowers generator degree and Bézout, exactly as the literature predicts, and does not touch the `d_reg(□)` growth. Characteristic 2, where the `Z/2` part is modular, is not covered.



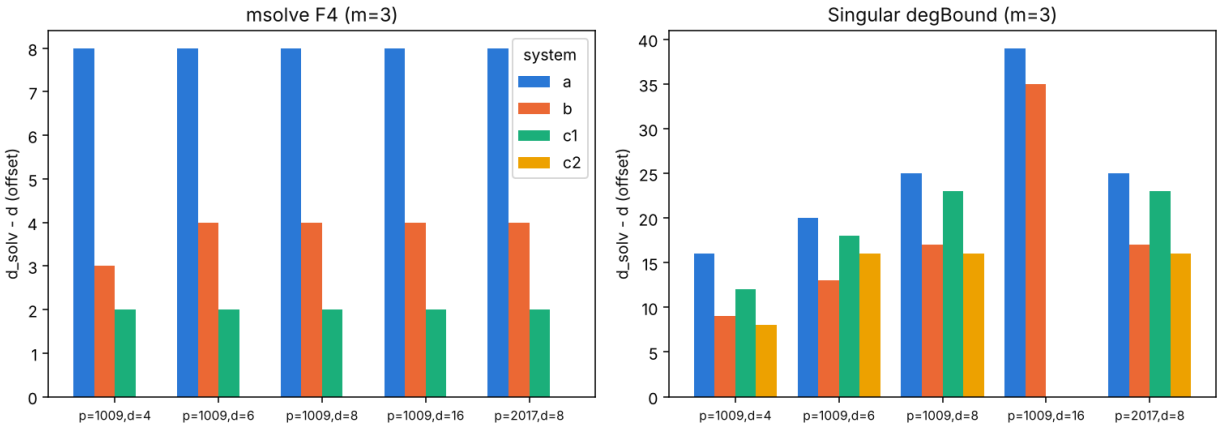
Rank 3: plain vs S_m vs D_m invariants at m=3 (log scale), both engines.

Rank 4 — Artin–Schreier trace elimination. A change to a trace-adapted basis leaves the ideal unchanged, and a closed-form polynomial elimination is impossible (the AS solution is rational in the factor-base coordinates and the validity trace `Tr(1/x)` is non-linear), so the residual enumerates the trace bits (2^k branches). The residual does lower the degree against the full system — but a **control** fixing an arbitrary non-trace linear form per unknown lowers it identically (27/27 Singular, 25/27 msolve, the other two ±1). The gain is generic variable fixing, not Artin–Schreier structure: **null**.



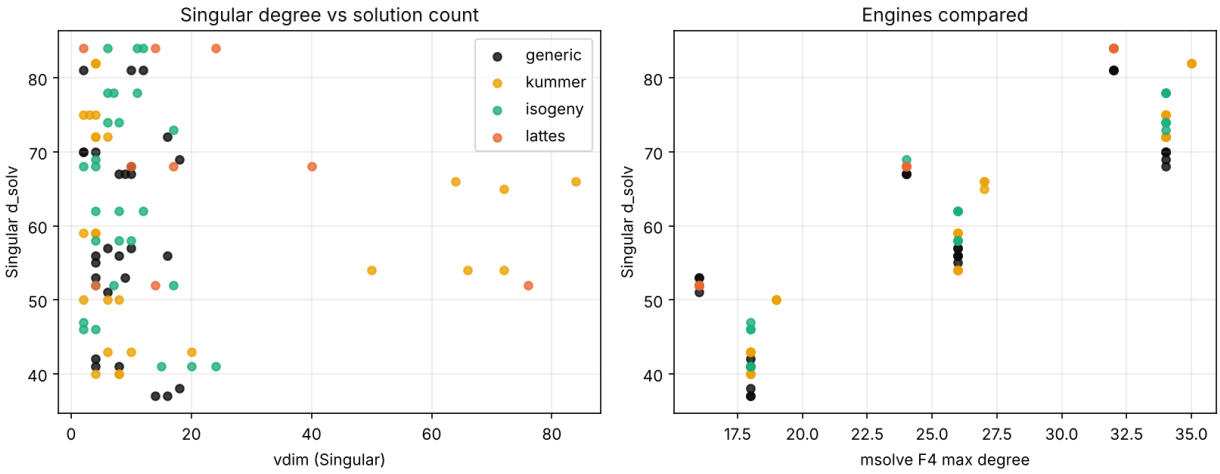
Rank 4: full vs Artin–Schreier residual vs non-trace control.

Rank 5 — a quadratic Kummer encoding. Introducing both `x(P+Q)` and `x(P−Q)` as unknowns (Vieta) and flattening to quadrics keeps every equation low-degree. On msolve the offset above the factor-base degree falls (chain 3–4, c1 2, c2 0), but c2 sits exactly at the input-degree floor F4 can never go below; on Singular the encoding is worse or mixed; Bézout rises 9–32× and the variable count grows 4→16. Consistent with the prior that algebraically dependent auxiliaries do not lower the intrinsic degree: **inconclusive, no usable lever.**



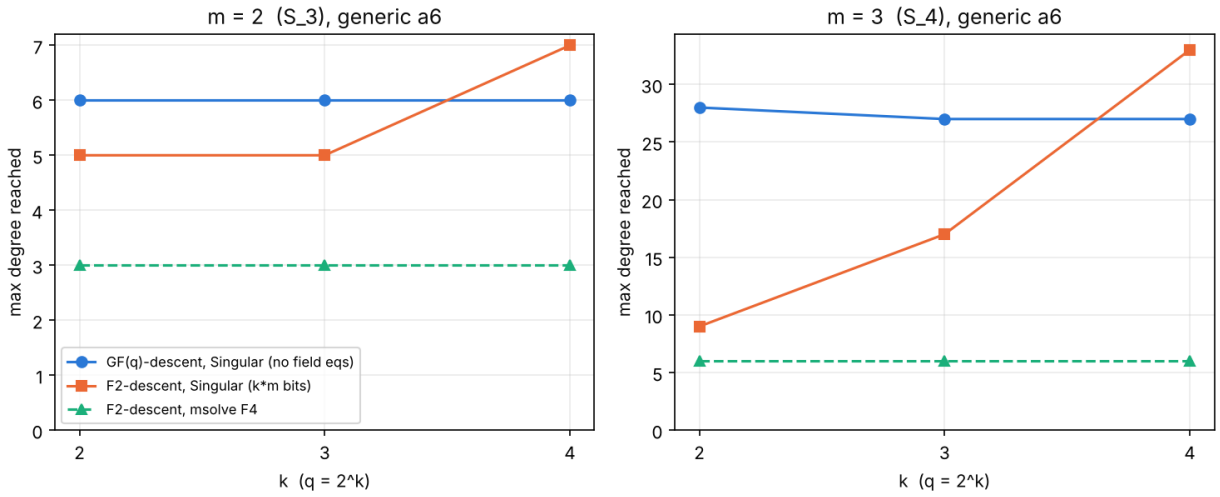
Rank 5: unsplit S4, S3-chain and two quadratic Kummer encodings.

Rank 6 — Lattès and isogeny factor bases. Over 90 systems the structured layer maps (Vélu 2-/3-isogeny chains, a Lattès `x([4])` map) give **exactly** the generic degree on msolve and slightly higher on Singular; decomposition yields are 1.01–1.14× a random base for every family, matching the program's earlier 0.93–1.03×. Caveat: in this design the degree is dominated by the base polynomial, so sensitivity to the layer structure is limited. **Null.**



Rank 6: generic, Kummer, isogeny and Lattès factor-base towers at equal degree.

Rank 7 — intermediate-field descent. Descending over `GF(2^k)` instead of `F_2` keeps equation degrees, Bézout and vdim identical across `k` and curves — Diem's bounded-degree regime, reproduced by the harness (a validation of the tooling against a theorem). Two honest limits: the contrasting growth of the full `F_2` descent shows on Singular but msolve stays flat at these small $k \le 4$; and a Koblitz curve with odd `m` has no generic planted decompositions (its abscissas collapse into `GF(q)`), so `m=3` used a non-Koblitz curve. For `n = 11, 13, 17, 131, 233, 283` there is no intermediate field: the lever is **structurally empty** for the target curves.



Rank 7: GF(q)-descent vs F_2-descent degrees across k.

What the eight exotic levers add up to. Four are null (ranks 1, 4, 6, 8), one is inconclusive (5), one validates the harness and is empty for prime `n` (7), and the two that lower the degree (2, 3) are both **symmetry quotients** — the one lever class the literature already identified, now measured on two engines. Rank 2 then shows that even a genuine per-decomposition degree drop can cost more in real linear algebra. None of the eight bends the `d_reg(□)` growth for binary Koblitz with prime `n` or for prime-Koblitz `j=0` curves.

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7. The named targets: binary Koblitz and prime-Koblitz (j=0)

Binary Koblitz / ECC2K-130 (prime `n`). Every degree lever above either is structurally empty (Frobenius subspaces: cyclotomic-coset argument), or lowers a constant without bending the `d_reg(□)` growth, or (fall reuse) holds degree 3–4 only until it saturates at `□=7` where the next degree is unmeasured for want of ~200 GB. The program's extrapolated ECC2K-130 cost rows span `2^68…2^197` and none drops below rho; the free-oracle floor `S·T^2 ≥ r` binds independently of the degree (T4).

Prime-Koblitz (`y^2=x^3+b`, `p≡1 mod 3`, j=0, order-6 automorphism). The order-6 automorphism `ζ:(x,y)↦(ωx,-y)` is currently used only as a GLV `÷6` fold on the factor base (`glv-orbit`), *not* as a symmetry quotient of the PDP ideal. The PKM-tower route (reproduced in §4.8) shows the prime-field solving degree is low (5–7 to N=18) but the cost is `29–141×` rho and the first arity that could win is `m=4`, needing an `S_5` small-root oracle at reach `δ≈1/5` — against a measured reach of `0.014` and a proven ceiling of `1/32`. The experiment v1 left open — using the order-6 automorphism as a symmetry quotient of the decomposition ideal itself — was run in v2 (§6.1, rank 2): it lowers the degree per decomposition for large factor bases, but its dense orbit polynomial makes the linear algebra 15–500× larger, so it is not a cheaper relation search. The automorphism remains best used as the existing `÷6` orbit fold.

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8. Open problems, what would unblock them, and the bottom line

Open problems carried forward (from the program's own records, with unblock criteria):

- `binary-boolean-system-64-variable-cap`: the system builder caps at 64 unknowns; 128 unknowns agreeing with the floor on 12 targets at `m=4,n=23,d=6` would unblock the `□=7` degree-5 measurement (currently out of memory at 169–405 GB).
- The **PKM / descent memory wall**: `m=3,N=21` needs a single ~192 GB process; **no problem record exists** for it (an AGENTS.md gap).
- The **inherited-F4 replay regression**: v1 called it a BMI2 host-path bug; v2 found the host paths already agree on `main` (since #2043) and that the real cause is a counted-work definition change (lazy → eager) after the sessions were recorded (§4.8). Now recorded as `problems/ecbench-claim-n13-replay-accounting-version-drift.md` (aburan28/crypto#2459). Unblock: an owner decision on which accounting is the method identity, plus an accounting-version field so cross-version replays are refused by name.
- **Magma never run** on the summation systems (T5): the "beat the fastest external tool" gate is formally unmet; an msolve-vs-Magma parity run on the PKM systems would close it.

The bottom line, now triple-checked. The degree of regularity of the Semaev PDP systems is not tamed by any tried lever: factor-base shaping and ordering are falsified (this report), symmetry/automorphism quotients lower constants and Bézout but not the `d_reg(□)` growth, fall-reuse holds a low *refutation* degree only up to a saturation point, and the semi-regular prediction overstates everything. For the named curves — binary Koblitz with prime `n`, and prime-Koblitz `j=0` — the structural levers are empty or uncompetitive, and **no end-to-end result is certificate-grade** (only component sub-facts are). What this study adds is not a break but the thing the program most needed: an **independent, twice-run, cross-engine measurement** confirming where the barrier is, why it is

intrinsic (CM-regularity invariance), and which of the field's open claims actually reproduce.

The single most valuable next action is not another lever but **making replay trustworthy**: wiring the validation sweep into the gate (done in v2, #2451) and making the counted-work definition version-stable (diagnosed in v2, #2459; needs the owner's lazy-vs-eager decision) — because an index calculus you cannot replay cannot be improved.

8.1 v2: the two reproducibility fixes

- **Validation sweep wired into the gate** — aburan28/crypto#2451. `load_verdicts()` now accepts the sweep's `row_id` (with the `A:/B:/C:` prefix normalised to the board slug) and recognises its vocabulary: `contradicted` fails and wins over everything; `validated` passes only with all three artifact pointers and no unclean caveat; `validated_with_caveats` / `unvalidated` are recognised but never passing, and are shown beside the row; an unknown verdict word stops the gate instead of being skipped. Effect: verdicts loaded 0 → 55 lines (79 keys); all 28 board rows stay "unvalidated (pre-gate)", now annotated with their sweep verdict and caveats — **deliberately**, since every one carries open caveats (e.g. `pricing_ok: false`). The gate was wired, not loosened. The single `contradicted` verdict is a diagnostic section, not a board row; nothing gates that section today, which is worth a follow-up.
- **Inherited-F4 counter drift** — aburan28/crypto#2459. The requested "BMI2 fix" turned out to be unnecessary: on `main` the BMI2 and portable rewrite paths already charge identical work (since #2043). The PR adds the regression test that proves it — both paths forced on the same random specialisation chains, every `InheritCost` field and row compared, failing if one charge is perturbed by 1 — and a problem record for the actual cause (§4.8). The charge itself was deliberately not changed: restoring the old lazy accounting would break every x86 session recorded since 2026-10-17 and still not fix the `ic-f4-m3` arm.

Appendix A — reproduce everything

All commands assume the harness directory. External tools: `msolve` 0.6.5, `Singular` 4.3.3.0. Deterministic outputs are noted; only timings vary.

```
# correctness certificates (prime + binary Semaev, ~2 min)
python3 cert_prime.py        # -> out/cert_prime.json   (0 FP / 0 FN)
python3 gf2.py               # -> out/cert_binary.json  (0 FP / 0 FN)
python3 cert_s4.py           # unsplit S4 certificate   (0 FP / 0 FN)

# determinism: run the canonical grid twice, diff (must be identical)
python3 canonical_run.py run1 ; python3 canonical_run.py run2
diff <(jq -S . out/canonical_run1.json) <(jq -S . out/canonical_run2.json)

# split vs unsplit, first-fall gap, engine comparison, V/order independence
python3 exp_split_vs_unsplit.py
python3 exp_engine_compare.py
python3 exp_fb_search.py --grid A        # subspace independence
python3 exp_order.py                     # order independence

# reproduce the program's PKM-tower msolve degrees (twice)
bash out/pkm/run_pkm.sh                   # all four match results.csv, both runs

# figures
python3 figures.py
```

Raw evidence (in `evidence/`): `cert_*.json`, `canonical_run{1,2}.json`, `split_vs_unsplit.json`, `engine_compare.json`, `exotic_fb_consolidated.json`, `exotic_order.json`, `REPRODUCTION_LOG.md`, the survey files in `surveys/`, and the exotic-lever directories `exotic/rank{2..7}_*/`.

Appendix B — provenance

- Program tree: `aburan28/crypto` @ `af663ad7c`; CLI: `aburan28/ecbench` (crypto pin `0dc9ef4`).
- Host: 4 vCPU, 15 GB RAM, Linux 6.18.44. `msolve` 0.6.5-1build2, `Singular` 4.3.3.0, `PARI/GP` 2.15.4, Python 3 with `sympy`, `numpy`, `matplotlib`, `reportlab`.
- Figures use the validated accessibility palette (light mode); every number traces to a file in `out/`.
- Surveys A/C/D/E were compiled by independent read-only passes over the program tree (survey C's inventory is partial — it stops at record C46 — but covers the split / root-finding / SAT slice used here); survey B was reconstructed directly after its agent was interrupted by a content filter. Every cross-referenced survey number was checked against the program's own files.